

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		All plots correct to $< \pm 1$ small square (2) 0,0 ignored 3 plots correct to $< \pm 1$ small square (1) Smooth curve from 0,0 to 45,48 through 0,0 (1) Penalise double lines / wispy lines		3		3	3	3
		(ii)	I	Volume (of hydrogen) increases as time increase (1) Until the volume produced is constant / at a decreasing rate / until it reaches 48 cm ³ / until it reaches 45 seconds (1)		2		2		2
			II	Read from graph (ecf) ± 1 small square Expect 36-38 cm ³		1		1	1	1
			III	45 s		1		1	1	1
	(b)	(i)		Increase {rate/speed } (of reaction) / lower activation energy Accept accelerate the reaction	1			1		
		(ii)		1.5 g	1			1		
		(iii)		Volumes (at each time) would be less	1			1		
	(c)	(i)		Volumes (at each time) would be greater	1			1		
		(ii)		Volumes (at each time interval) would be greater (but final collected volume would be the same) (1) Because particles move faster / collide more often (1)	2			2		
				Question 5 total	6	7	0	13	5	7